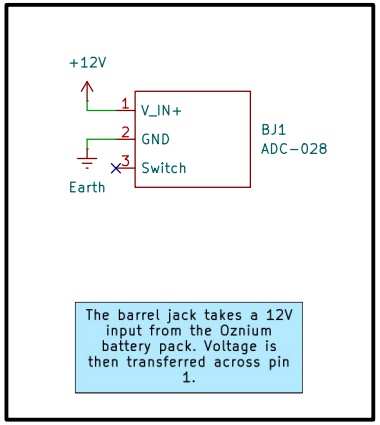
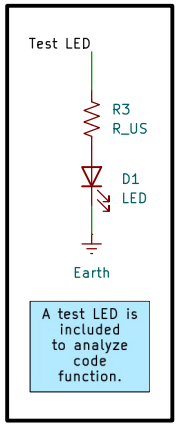
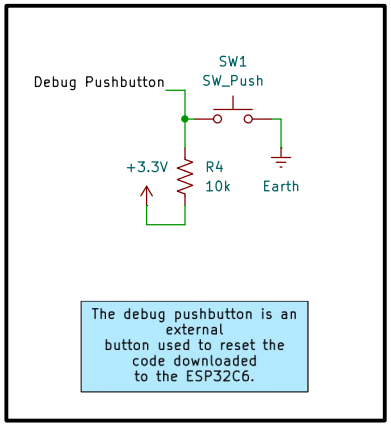
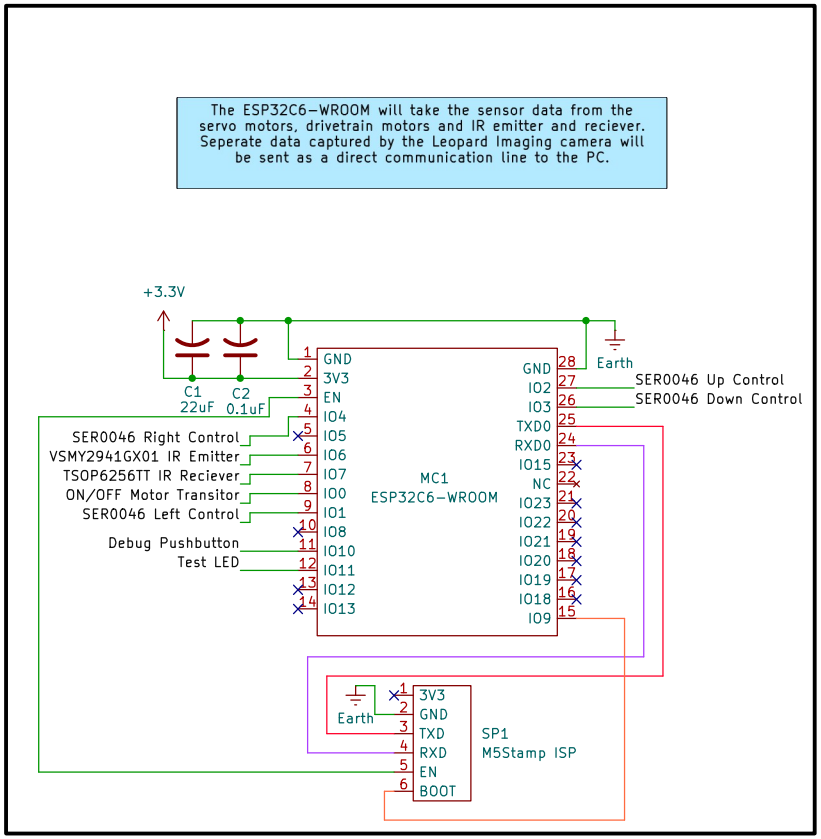
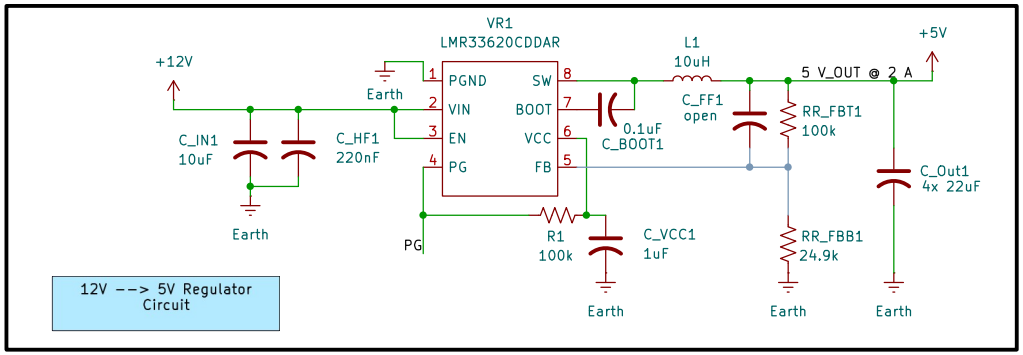
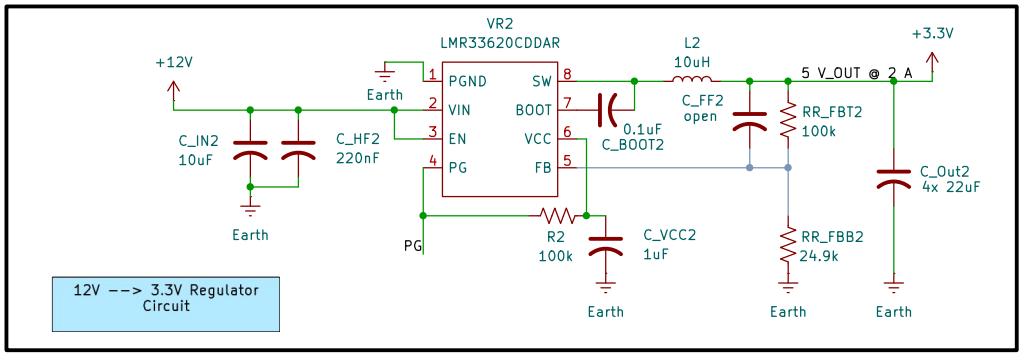

+12V
1
V_IN+
2
GND
3
Switch
BJ1
ADC-028
Earth
The barrel jack takes a 12V input from the Oznium battery pack. Voltage is then transferred across pin 1.


Test LED
R3
R_US
D1
LED
Earth
A test LED is included to analyze code function.


Debug Pushbutton
SW1
SW_Push
+3.3V
R4
10k
Earth
The debug pushbutton is an external button used to reset the code downloaded to the ESP32C6.


The ESP32C6-WROOM will take the sensor data from the servo motors, drivetrain motors and IR emitter and reciever. Seperate data captured by the Leopard Imaging camera will be sent as a direct communication line to the PC.
+3.3V
C1
22uF
C2
0.1uF
Earth
SER0046 Right Control
VSMY2941GX01 IR Emitter
TSOP6256TT IR Reciever
ON/OFF Motor Transitor
SER0046 Left Control
Debug Pushbutton
Test LED
GND
3V3
EN
IO4
IO5
IO6
IO7
IO0
IO1
IO8
IO10
IO11
IO12
IO13
MC1
ESP32C6-WROOM
GND
IO2
IO3
TXD0
RXD0
IO15
NC
IO23
IO22
IO21
IO20
IO19
IO18
IO9
SER0046 Up Control
SER0046 Down Control
3V3
GND
TXD
RXD
EN
BOOT
SP1
M5Stamp ISP


VR1
LMR33620CDDAR
+12V
+5V
Earth
PGND
VIN
EN
PG
SW
BOOT
VCC
FB
L1
10uH
5 V_OUT @ 2 A
C_IN1
10uF
C_HF1
220nF
0.1uF
C_BOOT1
C_FF1
open
RR_FBT1
100k
C_Out1
4x 22uF
R1
100k
C_VCC1
1uF
RR_FBB1
24.9k
PG
12V --> 5V Regulator Circuit


VR2
LMR33620CDDAR
+12V
+3.3V
Earth
PGND
VIN
EN
PG
SW
BOOT
VCC
FB
L2
10uH
5 V_OUT @ 2 A
C_IN2
10uF
C_HF2
220nF
0.1uF
C_BOOT2
C_FF2
open
RR_FBT2
100k
C_Out2
4x 22uF
R2
100k
C_VCC2
1uF
RR_FBB2
24.9k
PG
12V --> 3.3V Regulator Circuit